
SPECIALIZATION

VERSUS DIVERSIFICATION

AS A VENTURE CAPITAL

INVESTMENT STRATEGY

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EXECUTIVE SUMMARY

Much important work has informed us of rates of return earned by venture capitalists, the importance of venture capitalists to the "going public" process, and the criteria venture capitalists use to evaluate deals. This paper seeks to add to the literature by testing hypotheses, based upon both the finance and strategic management literature, regarding certain venture capitalist investment practices.

Venture capitalists seek to control or manage risk (Driscoll 1974; MacMillan, Siegel, and SubbaNarasimha 1985). Financing structure and investment strategy provide several means for venture capitalists to do this. Tools available to the venture capitalist include portfolio diversification to spread risk across different industries, firms, or hot/cold IPO markets to minimize unsystematic or investment-specific risk. Information sharing, networking, and specialization can also be used to control unsystematic risk.

Several hypotheses are developed from these conflicting perspectives. Data used to test the hypotheses are derived from responses to a survey of venture capitalists. Three hundred surveys were mailed to venture capitalists; 98, or 32.7%, returned usable responses.

Portfolio diversification is a well-known means to control risk exposure by reducing unsystematic or specific risks. However, Bygrave (1987, 1988), as well as financial intermediation theorists, argues that maintaining a high degree of specialization is useful for controlling risk as well as for gaining access to networks, information, and deal flow from other venture investors. The analyses of this paper build upon Bygrave's work. We construct more rigorous tests to resolve the conflict between the diversification and information-sharing hypotheses. Our hypothesis tests were usually resolved in favor of the information-sharing view. For example, venture capitalists in the sample that were heavily involved in seed round financing were diversified across fewer numbers of firms and industries.

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Further evidence in favor of information sharing is seen in investment patterns across different financing stages. Diversification would imply maintaining a portfolio of investments across the different investment stages. The information sharing/specialization view would argue that it is best to stay focused on a single stage or several "connected" stages. The empirical evidence from the sample once again favors the specialization perspective.

This research provides information of use to venture capitalists, as they seek information on how best to control risk; to entrepreneurs, as they learn of the factors venture capitalists consider in determining their investment strategy; and to academicians, as such studies provide insight to general industry practice and thus help to form the basis of classroom discussion and future research endeavors.

INTRODUCTION

As reported in Sahlman (1990), over one-third of investments by venture capitalists resulted in total absolute losses during the 1969–1985 time frame. About one of 15 investments results in returns that are ten or more times the venture capitalist's investment; the value of these few winners comprise about 49.4% of the ending value of the investor's portfolio and 61.4% of the investor's profits. The average holding period for a venture capital investment is 4.9 years, with about one-third of the investments held longer than six years. These statistics confirm that a successful outcome of a venture capital investment is highly uncertain and develops over a long time horizon. Ruhnka and Young (1991) have found that both the amount of risk and its composition varies across the stages of financing. In early stage financings, total risk is high and its major component comprises factors that are internal to the firm (e.g., team competence, technology development). In later stage financings, total risk declines and the major risk components are risk factors that are external to the firm (e.g., market response, competition, economic downturns).

This paper proposes and tests several hypotheses that deal with portfolio strategy as a means for venture capitalists to manage risk. The empirical results indicate that venture capitalists apparently use specialization and information-sharing strategies to control risk rather than financial diversification.

The following section reviews some means of risk control available to venture capitalists. Section 2 develops hypotheses regarding portfolio risk control. Section 3 describes the survey instrument used in the study. Section 4 reports the results of statistical tests. Concluding comments follow at the end of the paper.

1. RISK CONTROL

Previous studies have found that venture capitalists seek to manage and control risk (Driscoll 1974; MacMillan, Siegel, and SubbaNarasimha 1985). The screening process—in which only 1–3% of proposals receive funding—is certainly one potential means of controlling risk. Other means to control risk exist on both the "micro" or individual investment level and on the "macro" or portfolio level.

Micro Risk

Sahlman (1990) discusses that deal structure and active involvement by the venture capitalist in the portfolio firm are several "micro" methods of controlling risk. For example, rather than giving the entrepreneur a large lump sum of cash up front, most venture capital investments involve staged commitments of capital over time. Staging commitments allows the venture capitalist to maintain the valuable options to abandon, re-value, or expand their

investment in the firm. Injecting small increments of capital into the portfolio firm also helps to discipline the entrepreneurial team and keep them focused on developing a marketable product.

Many other deal structure provisions can seek to control “micro” elements of risk. The entrepreneurial team’s compensation contract can be structured so that cash salary is low but awards of stock options are high. Venture capitalists may be able to fire team members and repurchase their stock at less than market value. Venture capitalists may sit on the boards of their portfolio firms and be actively involved in the firm’s strategy setting and operations (Sahlman 1990; MacMillan, Kulow, and Khoylian 1989; Timmons 1987). Ratchet provisions, use of investment vehicles that are senior to the entrepreneur’s equity, put options, and redemption options are other means by which the venture capitalist can financially structure an investment in an attempt to control “micro” risk.

Unfortunately, research into various deal-specific risk control factors presents many problems, not the least of which is access to data. Due to their proprietary nature, contracts and term sheets between venture capitalists and entrepreneurs are not available for review. Information concerning general deal structure influences can be accessed through survey instruments (Norton and Tenenbaum 1992a, 1992b) but no “hard” data bases exist to provide this information to researchers. Additionally, whereas bond covenant provisions have language that has become fairly well standardized (Malitz 1986), this is probably not the case in the more fragmented and private venture capital industry. Thus an empirical study of deal structures that have developed to control “micro” risk is hindered by a lack of data availability and the potential complexity of the terms.

Macro Risk

“Macro” or portfolio level risk can also be controlled by the venture capitalist. Both financial theory and strategic management theory have developed perspectives for risk control in portfolios. Mainline finance theory argues that portfolios should contain a diversified collection of assets to minimize the effects of unsystematic or asset-specific risk (see, e.g., Sharpe 1964). However, strategic management theorists, as well as finance researchers who study financial intermediation, contend that portfolio specialization rather than diversification can be used to control portfolio risk (Bygrave 1987; Chan 1983; Campbell and Kracaw 1980). As with deal structure data availability, it is true that information on venture capitalist portfolio practices is not publicly available. However, a survey instrument can request basic information regarding the characteristics of venture capitalists’ portfolios. Statistical tests of survey responses can then determine if the sample data support the portfolio diversification or portfolio specialization theories of “macro” risk control.

2. DEVELOPING HYPOTHESES

Portfolio Diversification

Total risk (the variability in asset returns) has two components: systematic or market risk and unsystematic risk. Systematic risk arises from the effects of market or economy-wide influences on the returns to each asset. Unsystematic risk arises from firm, industry, or other asset-specific effects. By constructing a well-diversified portfolio, all unsystematic risk can be diversified away, leaving an investor exposed only to systematic risk influences. With a diversified portfolio, unfortunate circumstances that hurt returns on some assets are balanced by situations that increase returns on other assets.

Under the premises of the Capital Asset Pricing Model (CAPM), the financial markets reward only systematic risk with higher expected levels of return; exposure to unsystematic risk is not rewarded (Sharpe 1964). The arbitrage pricing theory (Ross 1976) has attracted a great deal of attention and empirical testing in the finance literature as a potential successor to the CAPM. It is a more general theory than the CAPM, but it still assumes that investors are fully diversified and subject only to systematic risk influences. Thus, finance theory generally assumes diversified investors. The only risk of interest is systematic or market risk; all firm or industry specific risk should be diversified away.

Venture capitalists are subject, *a priori*, to a great amount of unsystematic risk in each of their portfolio firms. Venture capitalists may want to diversify across different industries and companies in order to reduce their risk exposure in any one industry or product. Early stage investments carry great amounts of risk (Plummer 1987; Ruhnka and Young 1991). The desire to diversify may be especially strong for investors with large relative commitments to seed and first stage investments.¹ If venture capitalists seek to control unsystematic risk by way of portfolio diversification, the following hypothesis should be true:

H₁: Venture capitalists that make early stage investments will be more diversified to compensate for their risky individual investments. Thus, investors with greater relative commitments to early stage investments (i.e., seed or first stage) should be invested in relatively more industries and companies than those venture capitalists with lesser commitments to early financing stages.

Another source of asset-specific risk is liquidity risk. Ultimately venture capitalists wish to exit their investments and return either cash or liquid securities to their investment partners. Liquidity risk exists when the venture capitalist has difficulty exiting from an investment and receiving fair market value in return for the investment stake.

In the context of venture capital, there are two aspects to liquidity risk. The first aspect is deal-specific, i.e., the difficulty in exiting from an investment that is either failing or among the "living dead." This aspect of liquidity risk depends upon a great many factors, both internal and external to the entrepreneurial team. It can best be controlled, according to portfolio diversification theory, by constructing a well-diversified portfolio to reduce the effect of unsystematic risk on overall portfolio return.

The second aspect of liquidity risk deals with the timing of an exit from a successful investment. A venture capitalist would much rather seek to exit when the public equity and the acquisition/merger markets are placing fair values on portfolio firms. Returns to the venture capital investor may be lower if cashout occurs in markets that are "cold," or potentially less liquid. Thus, the portfolio diversification perspective implies that venture capitalists should not only seek to diversify across firms and industries but also across different financing stages. By staggering deals across different financing stages, some investments will be ready for cashout in "cold" initial public offering markets, some in "hot." The key is that the venture capitalist will be diversified in terms of cashout and be less likely to liquidate positions in substantial numbers of portfolio firms during cold exit markets. Thus, we have a second portfolio diversification hypothesis:

¹ It is the *relative* commitment to early stage investments that affects the investor's portfolio risk, not the absolute commitment. A venture capitalist with \$10 million in capital who is 80% invested in (undiversified) seed stage deals is *a priori* exposed to just as much unsystematic risk as a \$100 million pool that is 80% invested in (undiversified) seed deals.

H₂: Concerns about their inability to time “cold” and “hot” IPO markets will lead venture capital firms to diversify across different financing stages.

Specialization and Information Sharing

The CAPM includes such assumptions as homogenous investor expectations, no transactions costs, and equal access to information. These premises will not be true in the risk capital market. Thus the analysis of venture capitalists may be similar to those, such as financial intermediaries, that have information or transactions cost advantages over other investors; e.g., Leland and Pyle (1977), Campbell and Kracaw (1980), Chan (1983), and Bygrave (1987, 1988). Sahlman (1990) argues that valuable learning curve effects arise from the venture capitalist's specialized activity. The venture capitalist benefits from gaining institutional knowledge, sharing it with other investors and gatekeepers, earning a reputation for their experience and expertise, and is then able to “cultivate a deal flow based on networks of contacts and relationships” (Sahlman 1990, p. 500).

Financial theory has also developed models for specialized or concentrated investment activity. Levy (1978, 1991) and Merton (1987) construct models, called Generalized Capital Asset Pricing Models (GCAPM), which assume that segmented capital markets arise due to fixed costs of gaining information. As the reason for the segments disappear (i.e., as the fixed costs approach zero), the GCAPM equilibrium becomes identical to the CAPM as a special case. Levy (1978) argues that one source of fixed costs is the cost of discovering and interpreting the effects of events and information on specific firms in the investor's portfolio (e.g., reading and analyzing quarterly financial statements, keeping current on news and trends). In Levy's (1978) model, the existence of fixed costs limit the number of securities that appear in an investor's portfolio. Thus, complete diversification may not occur. Merton (1987) assumes investors possess specialized information that they then use to purchase stocks in specific areas. For example, a computer programmer should focus his/her investment activity in computer hardware and software firms; a physician will mainly invest in pharmaceutical and health-care firms. Under Merton's model, it is costly for investors to acquire information about companies that do business in areas unrelated to the investor's specialized expertise. Thus, the investor will invest only in those stocks about which they are already well-informed.

The assumption of complete diversification is not appropriate according to models that assume that some investors have cost advantages over others. From this perspective, venture capital firms will use their expertise to specialize in certain technical and product areas. Due to their information advantage in certain technologies or markets, and the high fixed costs of gaining expertise in other technical and product areas, it does not make economic sense for venture capitalists to seek portfolio diversification.

The knowledge base of venture capitalists include technological, market, and product expertise, as well as networks comprising experts and investors with similar interests. Venture capitalists seek to manage operating and technical risks by gaining access, by means of their reputation in their specialization, to information flows and deal flows in networks. According to Bygrave (1987, 1988), venture capitalists will invite other venture capitalists to participate in deals where their specialized knowledge is such that it will likely add value to the investment. We posit that venture capitalists that invest in firms involving the greatest amounts of technical and product risk (presumably early stage financing deals) should be more specialized, should have a more narrow industry focus, and may be less diversified than those who finance later stage deals.

Thus, if information sharing and specialization are a means to control risk, we have the following hypothesis to counter H₁:

H₃: Investors in early stage deals will have a narrow focus or industry specialization. Thus, the number of different industries and companies in which the venture capitalist is invested should be inversely related to the relative amount of seed and first stage investing done by the venture capitalist.

Diversification among the various financing stages will also be less important if specialization and networking theory is a better reflection of venture capitalist risk control practice. Venture capitalists will mainly invest in deals of a specific financing stage. Over time, the investor's financing stage profile may change somewhat as follow-on investments occur; but he/she will still seek to specialize in certain financing stages. Thus, the portfolio specialization and networking perspective leads to the following hypothesis to counter H₂:

H₄: The venture capitalist's strategy to specialize in order to enhance their position in networks and information flows will lead them to concentrate in one financing stage or several financing stages which may be related by virtue of subsequent follow-on investments.

Related Work

Bygrave (1987, 1988) reports on theoretical and empirical work related to the topic of this paper. Bygrave applies Pfeffer and Salancik's (1978) resource exchange model to the venture capital industry. The model predicts that the amount of networking or co-investing among venture capitalists should rise as uncertainty rises. In an analysis of the top 61 venture capital firms, Bygrave (1987) finds that the amount of co-investing is significantly greater among venture capitalists investing in high-tech firms than those investing in low-tech firms. He also discovers that the average investment per firm is significantly less for high-tech investments than for low-tech investments. Bygrave's results show that the amount of co-investing is not related to the size of the venture capitalist; small and large high-tech investors do similar amounts of co-investing, as do small and large low-tech investors. These results lead Bygrave (1987) to conclude:

Thus the principal reason for co-investing was not spreading of financial risk. Rather it was the sharing of expertise (p. 151).

Bygrave's test of the conflicting financial diversification and information sharing hypotheses is somewhat weak. For example, the fact that investors in high-tech firms did more co-investing and invested fewer dollars per portfolio firm than investors in low-tech firms is consistent with both the portfolio diversification and the information sharing viewpoints. The fact that the amount of co-investing was not related to the size of the venture capitalist does not disprove financial risk spreading among venture capitalists. Both large and small venture capitalists may have similar co-investing patterns if they have similar exposures to financial or technical risk.

The analysis to be done in this paper builds upon Bygrave's work. We seek to construct more rigorous tests to resolve the conflict between the diversification and information sharing hypotheses.

3. SURVEY STUDY

Surveys were mailed in February/March 1990 to 300 members of the National Venture Capital Association; a second mailing to non-respondents occurred in May/June 1990. A total of 98 responses were received for a response rate of 32.7%. By comparing mean responses

from the first and second mailings, the results suggest that response bias does not exist in the sample. Previous versions of the survey were pretested among eight venture capitalists. Their comments and suggestions were incorporated into the final survey format. No one who was part of the pre-test was included in the mailings. The cover letter promised the responses would be held in confidence. Data from the respondents is contained in Table 1.

The typical responding venture capital firm is a private limited partnership (item I).² In our sample, 75.5% of the respondents are limited partnerships; from data presented in Sahlman (1990), about 78–80% of the U.S. venture capital firms are organized as limited partnerships. Most of the respondents are larger funds, with over \$50 million of total funds managed (item IV). This is also in close agreement with Sahlman's (1990) industry statistics; he reports the average (median) capital size of an independent, private sector venture capitalist is \$65 million (\$30 million) in 1988. The portfolios of firms in our sample are fairly diversified; over half of the respondents' portfolios have investments in seven or more industries and nearly half have stakes in over 30 companies (items II and III).

The individual numerical responses on the percentage of financing done at each financing stage (item V) are summarized in the four columns presented in Table 1. The financing stages listed in item V are commonly used in the venture capital literature and are in standard use by practitioners (Sahlman 1990). The reported means are based upon the actual numerical responses given by the firms responding to the survey. Our sample appears to be more actively involved in seed stage investing than the industry averages reported in Sahlman (1990, table 1C). According to Sahlman's data, about 12.5% of venture capital funds are invested in seed stage investments, 67.5% in expansion stages, and 20.0% in leveraged buyouts. Our sample has about 24% of capital invested in seed stage deals, about 59% in expansion stage deals, and 17% in LBOs.

Comparing the size and composition of the venture capital firms in our sample to industry data reported in Sahlman (1990), the firms in our sample appear to be representative of firms in the industry. An exception is that the sample firms have relatively more capital invested in seed stage deals than the industry average. This should not make our sample invalid, as the focus of our analysis is on portfolio risk control. The greater relative amount of high-risk early stage investors in our sample may result in stronger tests of our hypotheses.³

4. HYPOTHESIS TESTING

Table 2, Panel A, reports correlation coefficients between the percentage of funds invested in early financing stages and diversification measures. Firms more actively involved in seed financing are diversified across a smaller number of industries and firms. This provides evidence against H_1 (financial risk diversification) and provides evidence in favor of H_3 (information sharing).

Contradictory findings occur when first round investments is examined. The number of industries in the portfolio is negatively related to the percentage of first round financing but the correlation coefficient is not significantly different from zero. The number of portfolio

² The statistical tests reported in this paper were done twice, once using the entire sample and once with only the responses from private limited partnerships. Qualitatively similar results occurred. This paper reports statistical tests using data from the entire sample.

³ It may be that the portfolio risk control strategies of "department store" venture capitalists differ from those of venture capital "boutiques." As the survey sample was not classified into one category or the other, the respondents are probably a mixture of both "department store" and "boutique" venture capitalists. Given that a number of our sample's characteristics correspond closely to Sahlman's (1990) "population" description, mixing venture capitalists of different investment philosophies should not adversely bias the results of our hypothesis tests.

TABLE 1 Aggregate Information Collected from Survey Items I through V

	Frequency
I. Type of venture capital firm:	
i. public	6
private	89
missing	3
ii. a. limited partnership	74
b. SBIC/MESBIC	10
c. corporate subsidiary	10
d. other	7
(some respondents checked more than one category)	
II. Number of different industries represented in the firm's portfolio	
1-3 industries:	10
4-6 industries:	25
7-9 industries:	28
10 or more:	33
missing:	2
III. Number of companies in the firm's portfolio	
1-9:	16
10-19:	20
20-29:	13
30 or more:	48
missing:	1
IV. Total amount of funds managed within the venture capital firm	
less than \$10 million:	9
\$11-20 million:	8
\$21-30 million:	9
\$31-40 million:	8
\$41-50 million:	3
over \$50 million:	59
missing:	2
V. Percentage of financing done at each stage by the respondents	

STAGE	PERCENTAGES				Average	Number of firms
	0-9%	10-20%	21-49%	50% or more		
start-up or seed	33	23	22	17	23.8%	95
first	25	27	32	10	22.1%	94
second	29	28	30	7	18.3%	94
third	46	24	18	6	13.1%	94
bridge	72	16	4	2	5.7%	94
other	60	9	11	14	17.1%	94

firms is positively and significantly related, at the 10% level, to the amount of first round financing. This provides weak evidence for H_1 , the financial risk diversification hypothesis.

Panel B of Table 2 reports the results of t -tests on differences in sample means. Venture capitalists reporting a large percentage of seed stage investing (50% of capital or more) were classified as the high seed group; those firms reporting less than 50% seed financing were placed in the low seed group. The high seed group is diversified across fewer industries and

TABLE 2 Relationship Between Early Stage Financing and Venture Capitalist Diversification

Panel A: Correlation coefficients between the percentage of funds invested in different financing stages and venture capitalist diversification measures. The probability values from a one-tailed test are in parentheses.

Early stage round:	Investor characteristics:	
	Number of industries in the portfolio	Number of companies in the portfolio
Seed	-.1861 (.037)	-.1666 (.054)
First	-.0446 (.336)	.1434 (.085)

Panel B: Results presented are *t*-ratios and, in parentheses, the probability values from a one-tailed test. The test seeks to determine if there is a relationship between venture capitalists with large early stage capital commitments and the diversification of their portfolio. The comparison in the *t*-test is between the number of different industries and firms in the portfolios of those venture capital firms with 50% or more of their funds invested in the listed stage against firms that have fewer than 50% of their funds invested in the designated financing stage. A negative *t*-ratio implies investors with 50% or more of their funds in the listed stage have less-diversified portfolios than those investors that have fewer than 50% of their funds invested in the listed stage.

Early stage round:	Investor characteristics:	
	Number of industries in the portfolio	Number of companies in the portfolio
Seed:	-1.49 (0.07)	-2.21 (.015)
First:	-1.74 (.043)	-1.92 (.030)

has investments in a smaller number of companies than the low seed group. The respondents were similarly grouped according to their percentage of first round investments. The venture capitalists that have a large percentage of first round investments also are less diversified across industries and companies, in support of H_3 .

Though some conflicting results occur when correlation coefficients are examined, the results of *t*-tests that focus on investors with large relative commitments to early stage investments indicate strong evidence in favor of the information sharing hypothesis, H_3 . These tests provide some empirical evidence that, among the venture capitalists in the sample, the management of technical, marketing, and other firm-specific risks via specialization is of greater importance than managing financial risk.⁴

Table 3 shows the correlation matrix of the percentage of funds invested in the different financing stages. Diversification and H_2 would be supported if no pattern of significant correlations occurred; e.g., a venture capitalist's involvement in first stage financing should have no impact on the degree of second stage investment.

⁴These results are not being driven by smaller venture capitalists. Venture capitalists across all size categories (item IV in Table 1) occur across all diversification measures (number of industries, number of companies in the portfolio) and financing stages.

TABLE 3 Relationship Between the Percentage of Funds Invested by Venture Capitalists in Each Financing Stage

	First	Second	Third	Bridge	Other
Seed	.0318 (.381)	-.2904 (.002)	-.3758 (.000)	-.1713 (.049)	-.3857 (.000)
First		.1368 (.094)	-.3865 (.000)	-.2521 (.007)	-.3743 (.000)
Second			.0347 (.370)	-.1019 (.164)	-.4025 (.000)
Third				.1862 (.036)	-.1906 (.033)
Bridge					-.1780 (.043)

Note: Correlation coefficients between the percentage of venture capitalist funds invested in the various financing stages. Correlations are based upon actual numerical responses given in the survey. Probability values from a one-tailed test are in parentheses.

Evidence of related investment patterns would indicate support for H_4 (information sharing). If venture capitalists specialize, the percentage of financing in a given stage may be positively related to those of nearby stages as a result of follow-on investments. In addition, a negative relationship should exist between financing stages that are not consecutive. For example, whereas seed and first round investing may be positively correlated as a result of follow-on investments, seed and third stage financing should be negatively related.

The latter is what appears in Table 3. For the most part, positive correlations exist between successive financing stages (as seen on the diagonal).⁵ This implies some degree of follow-on investment activity occurs in the firms' portfolios. Off-diagonal correlations, which represent non-connected financing stages, are mostly negative and statistically significant. The evidence from Table 3 supports the specialization hypothesis, H_4 . No support of the portfolio diversification hypothesis, H_2 , is indicated.

5. CONCLUSIONS

This paper reviews responses to a survey completed by 98 venture capitalists. The paper seeks to determine if venture capitalists attempt to control risk through competing portfolio strategies. We examine various sources of risk and how strategies to manage risk can affect portfolio construction.

Portfolio diversification is a method to reduce risk exposure by financial means. Venture capitalist portfolios can reduce unsystematic risk by investing in a diversified set of firms or industries. The portfolios can also be diversified against the risk of liquidation during cold IPO markets by containing investments in different stage deals.

Bygrave's (1987, 1988) work on the role of networks, specialization, and information sharing argues otherwise. To control risk, venture capitalists should seek to specialize in order to exploit their technical and product expertise. The specialization hypothesis implies venture capitalists will have portfolios that are less diversified across companies, industries, and financing stages.

⁵ "Other" stage financing involves mainly leveraged buyout and management buyout deals in our sample. Thus, they should not be considered as a follow-on investment to a "bridge" deal.

The survey data allow for several statistical tests of conflicting hypotheses from the diversification and networking perspectives. Most of the conflicts were resolved in favor of the specialization and information sharing theory. Evidence favoring the specialization hypothesis includes the finding that investors in seed deals are less diversified across different industries and firms. Another finding is that venture capitalists appear to specialize in certain financing stages rather than stagger their investments over different financing stages.

This paper reports fairly robust empirical evidence in favor of the perspective that venture capitalists control portfolio risk through their efforts to specialize, to build reputation capital, and to become important members of information and deal flow networks.

Additional work is needed to better understand the theory and process of venture capitalist risk control. This paper has examined risk management from the "macro," or overall portfolio perspective. Research is also needed on risk management at the "micro," or individual investment, level, although such studies are hindered by lack of access to data. It is also of interest to examine the interrelationship, if any, between risk management at the macro and micro levels. Also, why aren't venture capitalist "supermarkets" more prevalent? Venture capitalist supermarkets can allow the best of both worlds—an overall portfolio diversified in terms of time and industry, with experts overseeing sub-portfolios that are specialized in terms of industries and financing stage.

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